

MACSE622 – DATA ENGINEERING AND VISUALIZATION LAB

Register Number	25MCB1010	Subject Code	MACSE622
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Programme	MTech CSE (Big Data Analytics)	Slot	L33 + L34
Course	Data Engineering and Visualization Lab	Semester	2nd Semester
Date	02-02-2026	Exp No	6

Ex No: 6

Aim: Program an Apache Airflow ETL

Implementation:

1. **source** ~/airflow_venv/bin/activate
2. In ~/airflow/dags/ create python file: **nano** [tasks.py](#)
3. **airflow standalone**
4. Open <https://localhost:8080>

Code ([tasks.py](#)) :

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from airflow.operators.empty import EmptyOperator
from datetime import datetime
```

```
def task_1(**context):
    a = 10
    b = 15
    ti = context["ti"]
    ti.xcom_push(key="a", value=a)
    ti.xcom_push(key="b", value=b)
```

```
def task_2(**context):
    ti = context["ti"]
```

```

a = ti.xcom_pull(key="a", task_ids="task_1")
b = ti.xcom_pull(key="b", task_ids="task_1")
total = a + b
ti.xcom_push(key="sum", value=total)

```

```

def task_3(**context):
    ti = context["ti"]
    total = ti.xcom_pull(key="sum", task_ids="task_2")
    if total % 2 == 0:
        print(f"{total} is even")
    else:
        print(f"{total} is odd")

```

```

with DAG(
    dag_id="three_task_linear_dag",
    start_date=datetime(2026, 2, 2),
    schedule="@daily",
    catchup=False,
) as dag:

```

```

    start = EmptyOperator(task_id="start")

```

```

    t1 = PythonOperator(
        task_id="task_1",
        python_callable=task_1,
    )

```

```

    t2 = PythonOperator(
        task_id="task_2",
        python_callable=task_2,
    )

```

```

    t3 = PythonOperator(
        task_id="task_3",
        python_callable=task_3,
    )

```

```

    end = EmptyOperator(task_id="end")

```

```

    start >> t1 >> t2 >> t3 >> end

```

Airflow page (<https://localhost:8080>)

The screenshot shows the Airflow web interface at localhost:8080. The left sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main content area displays the 'Task Instances' tab for the DAG 'three_task_linear_dag' with the run ID 'manual__2026-02-02T11:16:23+00:00'. The run status is 'Success'. A table lists the task instances:

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
start		Success	2026-02-02 16:46:26	1	EmptyOperator	00:00:00.068
task_1		Success	2026-02-02 16:46:27	1	PythonOperator	00:00:01.228
task_2		Success	2026-02-02 16:46:28	1	PythonOperator	00:00:01.228
task_3		Success	2026-02-02 16:46:29	1	PythonOperator	00:00:00.058
end		Success	2026-02-02 16:46:30	1	EmptyOperator	

DAG:

The screenshot shows the Airflow web interface at localhost:8080. The left sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main content area displays the DAG canvas for the DAG 'three_task_linear_dag' with the run ID 'manual__2026-02-02T11:16:23+00:00'. The DAG is a linear flow with the following tasks:

```
graph LR; start[EmptyOperator] --> task_1[PythonOperator]; task_1 --> task_2[PythonOperator]; task_2 --> task_3[PythonOperator]; task_3 --> end[EmptyOperator];
```

The tasks are all in a 'Success' state. The right sidebar shows the 'Task Instances' tab for the DAG run, listing the task instances and their states.

Input of task 1 is 10 and 15.
Output of task 2 is $10+15 = 25$
Output of task 3 is "25 is odd"

Task 1: Audit logs

The screenshot displays the Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, and Admin. The main area shows a DAG named 'three_task_linear_dag' with a 'Dag Run' status of 'manual_2026-02-02T11:26:39+00:00'. The task 'task_1' is highlighted. The right-hand panel shows the 'task_1' details, including its operator 'PythonOperator', start date '2026-02-02 16:56:42', end date '2026-02-02 16:56:42', and duration '00:00:00.061'. The 'Logs' tab is selected, showing a list of log messages with their timestamps and sources. The log messages include information about DAG bundles, task instance state transitions, and task completion.

task_1 Success Add a note Clear Task Instance Mark as Failed

Operator	Start Date	End Date	Duration
PythonOperator	2026-02-02 16:56:42	2026-02-02 16:56:42	00:00:00.061

Logs | Rendered Templates | XCom | Asset Events | Audit Log | Code | Details

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```
Log message source details sources=["/home/student/airflow/logs/dag_id=three_task_linear_dag/run_id=manual_2026-02-02T11:26:39+00:00/task_1"]
2 [2026-02-02 16:56:42] INFO - DAG bundles loaded: dags-folder, example_dags source=airflow.dag_processing.bundles.manager
3 [2026-02-02 16:56:42] INFO - Filling up the DagBag from /home/student/airflow/dags/tasks.py source=airflow.models.dagbag
4 [2026-02-02 16:56:42] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.operators.empty.EmptyOperator' instead.
5 [2026-02-02 16:56:42] WARNING - The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.operators.empty.EmptyOperator' instead.
6 [2026-02-02 16:56:42] INFO - Task instance is in running state source=task.stdout
7 [2026-02-02 16:56:42] INFO - Previous state of the Task Instance: TaskInstanceState.QUEUED source=task.stdout
8 [2026-02-02 16:56:42] INFO - Current task name: task_1 source=task.stdout
9 [2026-02-02 16:56:42] INFO - Dag name: three_task_linear_dag source=task.stdout
10 [2026-02-02 16:56:42] INFO - Done. Returned value was: None source=airflow.task.operators.airflow.providers.standard
11 [2026-02-02 16:56:42] INFO - Task instance in success state source=task.stdout
12 [2026-02-02 16:56:42] INFO - Previous state of the Task Instance: TaskInstanceState.RUNNING source=task.stdout
13 [2026-02-02 16:56:42] INFO - Task operator: <Task(PythonOperator): task_1> source=task.stdout
```

Task2: Audit logs

The screenshot displays the Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, and Admin. The main area shows a DAG named 'three_task_linear_dag' with a 'Dag Run' status of 'manual_2026-02-02T11:26:39+00:00'. The task 'task_2' is highlighted. The right-hand panel shows the 'task_2' details, including its operator 'PythonOperator', start date '2026-02-02 16:56:43', end date '2026-02-02 16:56:43', and duration '00:00:00.049'. The 'Logs' tab is selected, showing a list of log messages with their timestamps and sources. The log messages include information about DAG bundles, task instance state transitions, and task completion.

task_2 Success Add a note Clear Task Instance Mark as Failed

Operator	Start Date	End Date	Duration
PythonOperator	2026-02-02 16:56:43	2026-02-02 16:56:43	00:00:00.049

Logs | Rendered Templates | XCom | Asset Events | Audit Log | Code | Details

All Log Levels ▼ All Sources ▼ Log Settings

```
Log message source details sources=["/home/student/airflow/logs/dag_id=three_task_linear_dag/run_id=manual_2026-02-02T11:26:39+00:00/task_2"]
2 [2026-02-02 16:56:43] INFO - DAG bundles loaded: dags-folder, example_dags source=airflow.dag_processing.bundles.manager
3 [2026-02-02 16:56:43] INFO - Filling up the DagBag from /home/student/airflow/dags/tasks.py source=airflow.models.dagbag
4 [2026-02-02 16:56:43] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.operators.empty.EmptyOperator' instead.
5 [2026-02-02 16:56:43] WARNING - The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.operators.empty.EmptyOperator' instead.
6 [2026-02-02 16:56:43] INFO - Task instance is in running state source=task.stdout
7 [2026-02-02 16:56:43] INFO - Previous state of the Task Instance: TaskInstanceState.QUEUED source=task.stdout
8 [2026-02-02 16:56:43] INFO - Current task name: task_2 source=task.stdout
9 [2026-02-02 16:56:43] INFO - Dag name: three_task_linear_dag source=task.stdout
10 [2026-02-02 16:56:43] INFO - Done. Returned value was: None source=airflow.task.operators.airflow.providers.standard
11 [2026-02-02 16:56:43] INFO - Task instance in success state source=task.stdout
12 [2026-02-02 16:56:43] INFO - Previous state of the Task Instance: TaskInstanceState.RUNNING source=task.stdout
13 [2026-02-02 16:56:43] INFO - Task operator: <Task(PythonOperator): task_2> source=task.stdout
```

Task3: Audit logs

The screenshot displays the Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, and Admin. The main area shows a DAG named 'three_task_linear_dag' with a 'Dag Run' status of 'manual_2026-02-02T11:26:39+00:00'. The DAG consists of three tasks: 'task_1' (PythonOperator, success), 'task_2' (PythonOperator, success), and 'task_3' (PythonOperator, success). The 'task_3' details panel is open, showing the operator as 'PythonOperator', start date as '2026-02-02 16:56:44', end date as '2026-02-02 16:56:44', and duration as '00:00:00.038'. The 'Logs' tab is selected, displaying a list of log messages. The log messages include information about DAG bundles, task instance state, and task completion.

task_3 Success

Operator: PythonOperator Start Date: 2026-02-02 16:56:44 End Date: 2026-02-02 16:56:44 Duration: 00:00:00.038

Logs Rendered Templates XCom Asset Events Audit Log Code Details

All Log Levels: ▼ All Sources: ▼ Log Settings: ⚙️

```
2 [2026-02-02 16:56:44] INFO - DAG bundles loaded: dags-folder, example_dags source=airflow.dag_processing.bundles.manu
3 [2026-02-02 16:56:44] INFO - Filling up the DagBag from /home/student/airflow/dags/tasks.py source=airflow.models.dag
4 [2026-02-02 16:56:44] WARNING - The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'ai
5 [2026-02-02 16:56:44] WARNING - The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'ai
6 [2026-02-02 16:56:44] INFO - Task instance is in running state source=task.stdout
7 [2026-02-02 16:56:44] INFO - Previous state of the Task Instance: TaskInstanceState.QUEUED source=task.stdout
8 [2026-02-02 16:56:44] INFO - Current task name:task_3 source=task.stdout
9 [2026-02-02 16:56:44] INFO - Dag name:three_task_linear_dag source=task.stdout
10 [2026-02-02 16:56:44] INFO - 25 is odd source=task.stdout
11 [2026-02-02 16:56:44] INFO - Done. Returned value was: None source=airflow.task.operators.airflow.providers.standard
12 [2026-02-02 16:56:44] INFO - Task instance in success state source=task.stdout
13 [2026-02-02 16:56:44] INFO - Previous state of the Task Instance: TaskInstanceState.RUNNING source=task.stdout
14 [2026-02-02 16:56:44] INFO - Task operator:<Task(PythonOperator): task_3> source=task.stdout
```

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